DRISTII PATEL

BS (Research) Physics · Indian Institute of Science (IISc), Bengaluru

Neutrino physics · Particle detectors · Computational astroparticle physics

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SUMMARY

Third-year BS (Research) Physics student at IISc working with Dr. Biswaranjan Behera on supernova and solar neutrino detection in deuterated liquid scintillators. Re-implemented four published analyses from scratch, built detector-rate and sensitivity pipelines that extend them to new detector configurations and sterile-neutrino scenarios, and am contributing a diffuse supernova neutrino background (DSNB) module to the open-source SNEWPY package. A manuscript on solar ^{8}B neutrino sensitivity to sterile-neutrino scenarios is in preparation.

EDUCATION

Bachelor of Science (Research), Physics 2024 – 2028
Indian Institute of Science (IISc), Bengaluru, India (NIRF 2025: #1 University · QS World 2027: #221)

- **CGPA 8.3/10**; fourth-semester TGPA 9.0/10
- **Admission:** JEE Advanced 2024, **AIR 1925** (of ∼180,000 candidates); JEE Main 2024, AIR 3898 (99.76 percentile)
- **Coursework:** Intermediate Classical Mechanics; Intermediate Electromagnetism & Quantum Physics of Radiation; Intermediate Thermal Physics & Materials; Introduction to Quantum Measurements

Higher Secondary (GSEB), Crystal School 86.78% (Physics 91, Mathematics 96) 2022 – 2024

RESEARCH EXPERIENCE

Undergraduate Researcher | Supernova & Solar Neutrino Detection 2025 – Present
Centre for High Energy Physics, IISc | Supervisor: Dr. Biswaranjan Behera (SERB Ramanujan Fellow)

Computational modelling of supernova and solar neutrino signals in deuterated liquid scintillator (DLS) detectors: event-rate calculations, detector-response modelling and sensitivity analysis.

Reproduction of DLS supernova-neutrino event rates (Chauhan, Dasgupta & Datar, JCAP 2021) [code]

- Rebuilt the event-rate pipeline from the published description, without the original code (fluence, digitised NSGK deuteron cross sections, Birks'-law quenching, CC/NC/ES spectra); **NC rates within 1–3%** of the published tables
- Investigated an initial 17–20% CC discrepancy: ruled out coding errors with an independent NumPy re-implementation, traced it to coarse sampling in the printed cross-section tables, and resolved it to <**4%** using the original NSGK tables
- Extended beyond the paper: recoil-neutron spectra from EXFOR data, secondary breakups, deuteron secondary scintillation

Neutrino mass-ordering sensitivity across eight detector configurations [code]

- Developed a Poisson-significance framework with adiabatic MSW conversion to compare mass-ordering sensitivity across eight detectors (H_2O, D_2O, WbLS, Gd/Te-doped, D_2O+LAB) in IBD, CC, NC and ES channels; D_2O rates agree with published values to ∼1%
- Used time-dependent SNEWPY fluxes (Nakazato, Bollig, Tamborra) with adaptive time-binning to recover accretion-phase sensitivity lost in whole-burst integration; **identified the choice of supernova simulation as the dominant source of sensitivity variation**

Solar ^{8}B neutrino survival probability: sterile-neutrino sensitivity [code]

Manuscript in preparation

- Computed $P_{ee}(E)$ for three-flavour LMA-MSW and 3+1 sterile-neutrino models by numerically evolving the Hamiltonian through realistic solar density profiles; identified the energy window of maximum model separation and the exposure needed to discriminate them
- Day–night asymmetry with the PREM Earth density profile; CC spectra and energy-dependent asymmetry for four DLS configurations in a 1 kton detector

OPEN-SOURCE SOFTWARE

SNEWPY (SNEWS2 Collaboration) : Diffuse Supernova Neutrino Background (DSNB) Module 2026 – Present
*Python, NumPy, SciPy · github.com/SNEWS2/snewpy · **pull request under review***

- Implemented redshift-dependent DSNB flux calculations from cosmological core-collapse rates and IMF-weighted progenitor spectra, with per-progenitor MSW flavour transformation supporting both mass orderings; validated against published DSNB calculations
- Integrated with SNEWPY's supernova-model, Flux/Fluence, flavour-transformation and RateCalculator APIs, avoiding duplicated cross-section code; **maintained a 21-test suite through iterative refactoring and maintainer code review**

INDEPENDENT RESEARCH PROJECTS

DSNB Detection Sensitivity: Reproduction of Li, Vagins & Wurm (2022) 2025 – 2026
Python, NumPy, SciPy, Matplotlib · gitlab.com/dristiipatel/dsnb-reproduction

- Reproduced Table III (SK-Gd and JUNO) **to rounding precision** from first principles (arXiv:2201.12920): DSNB flux over the evolving core-collapse rate, Strumia–Vissani IBD cross section, detector and background models; vectorised the 2D (E, z) integration for a $\sim 50\times$ speed-up
- Produced five figures beyond the paper, including a 2D significance map in $(\langle E_\nu \rangle, f_{\mathrm{BH}})$ space; documented the remaining $\sim$15% signal-rate discrepancy with the published calculation rather than correcting it empirically

CID Method Simulation: Directional Solar-Neutrino Extraction (Borexino) 2026
Python, NumPy, SciPy, Matplotlib · gitlab.com/dristiipatel/cid-solar-neutrino-sim

- Simulated Borexino's Correlated and Integrated Directionality technique (Basilico et al., arXiv:2307.14636) with a synthetic PMT-sphere detector (scintillation and directional Cherenkov hits); joint χ^2 fit across Nth-hit $\cos\alpha$ slices with a Bayesian posterior on signal counts
- Calibration check: 68% credible intervals covered the true value in 73% of 30 trials; simplifications relative to the published analysis documented

GW150914 via Matched Filtering 2026
Python, PyCBC, GWOSC · gitlab.com/dristiipatel/gw-matched-filter

- Built the pipeline (GWOSC data, PSD estimation, whitening, IMRPhenomD templates); recovered GW150914 independently in H1 and L1 with peak SNR **17.55** and **13.10**, within $\sim$30 ms of the published merger time

TRAINING & DIRECTED READING

Summer School for Women in Physics (SSWP), ICTS Bengaluru June 2025
Intensive research programme

- Experimental methodology and uncertainty analysis; laser-diffraction and Brownian-motion experiments with extracted diffusion coefficients

High Energy Physics & Particle Detectors March 2026
Directed reading project under Dr. Behera, IISc

- Radiation detection, charged-particle energy loss and Cherenkov radiation (Knoll; Garutti's HEP detector lecture notes)

SCIENCE COMMUNICATION

IISc Open Day 2026 : Live Muon Detection Demonstration March 2026
Indian Institute of Science, Bengaluru · public audience

- Set up and operated a live muon-detection experiment, explaining Cherenkov radiation and charged-particle detection to visitors without a physics background

HONORS & AWARDS

Deep Asher Fellowship 2024–25 (IISc Class of 1998 ME(Int)) : one B.Sc student per year, awarded on entrance-exam rank and first-semester TGPA 2024

NSEA 2024 (IAPT) : state-level selection for the Indian National Astronomy Olympiad (INAO) 2024

IAAC 2025 : Round 2 qualifier; Round 1 submission featured on the official website for clarity of reasoning 2025

LABORATORY EXPERIENCE

30+ experiments at IISc, with Python for data processing and graph generation throughout, including Cornu's method, Mach–Zehnder interferometry, Fabry–Perot resonator, ultrasonic diffraction, electron diffraction, Compton scattering, atomic spectra (Rydberg constant), Zeeman effect, Nd:YAG laser, Hall effect, Faraday effect, B–H hysteresis, Seebeck & Peltier effects, Wiedemann–Franz law, Stirling engine, semiconductor I–V and microwave optics.

SKILLS

Scientific computing: Python, NumPy, SciPy, Matplotlib · **Research software:** SNEWPY, PyCBC, SNOwGLoBES · **Tools:** Git, GitHub, GitLab, LaTeX

Methods: numerical modelling, event-rate calculations, sensitivity analysis, parameter scans, uncertainty analysis, reproduction and validation of published results

Languages: English (professional), Hindi (native), Gujarati (native)

ACTIVITIES & INTERESTS

Ensemble (IISc Physics Club); Samasya (IISc Maths Club); Pravega 2025 event coordination; Sangam performances (keyboard, vocals). *Interests:* keyboard, singing, sketching, journaling.

REFERENCE

Dr. Biswaranjan Behera : research supervisor; SERB Ramanujan Fellow, Centre for High Energy Physics, IISc